Harry Nguyen

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Education

Georgia State University Expected May 2027
Bachelor of Science in Computer Science · **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

Work Experience

Google | Software Engineer Intern — Chrome Browser May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and cross-language compatibility
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven **TypeScript/React** system with observer pattern across **25K+ lines** of Chromium at **95% test coverage**; design documents adopted into production branch by senior engineers

Develop for Good | Software Engineer Intern May 2024 – Aug 2024

- Designed stateless BaaS on AWS — selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

CoderPush | Software Engineer Intern Sep 2023 – Dec 2023

- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy — **30% read throughput improvement** under sustained high-concurrency traffic
- Designed idempotent payment APIs with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

Personal Projects

TiMoto AI — ML Serving & Evaluation Platform | *vLLM, gRPC, Python, Django, AWS, Terraform* timoto.ai

- **Sole engineer** — designed, built, and operate end-to-end AI evaluation platform: **vLLM** inference engine, gRPC inter-service layer, **Python**/Django REST backend, PostgreSQL, and multi-AZ AWS infrastructure
- Selected **vLLM with PagedAttention** over naive inference to eliminate KV cache fragmentation under concurrent load; deployed with continuous batching and **LLM-as-a-judge evaluation pipeline** — zero OOM failures at production traffic
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause to circular resource acquisition; redesigned call sequencing — **100% evaluation success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with Terraform IaC and circuit breaker pattern — **99.9% uptime**, **44% cost reduction** to $40–60/month; auto-rollback on health check failure

Pulumi — Open Source Contributor | *Go, TypeScript, IaC* 24.4K★ github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

Technical Skills

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, LoRA/QLoRA (PEFT), RAG pipelines, PyTorch, LangChain, Ray
Languages: Python, TypeScript, C++, Go, JavaScript, Java, SQL, Bash, Rust
Frameworks & Databases: React, Django, FastAPI, Node.js, REST APIs, PostgreSQL, Redis, MongoDB, DynamoDB
Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions
Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Raft/Paxos, Fault tolerance